

Evaluating Classifier Performance

Albert Y. Kim

Motivating Example

- ▶ [Sobar \(2016\)](#) aimed to predict cases of cervical cancer using non-invasive social and behavioral assessments
 - ▶ They recorded 18 different assessment scores for 72 individuals, 21 with cervical cancer and 51 without cancer
- ▶ A k -nearest neighbors model (using $k = 8$) evaluated using LOOCV results in 87.5% classification accuracy
 - ▶ Is this a worthwhile result?

Confusion Matrices

The table below displays the actual status and the cross-validated prediction of all 72 individuals:

	Predicted Cancer	Predicted Healthy
Has Cervical Cancer	13	8
Doesn't Have Cancer	1	50

Despite being $< 30\%$ of the data, individuals with cancer made up 8 of 9 incorrect classifications.

Confusion Matrices

The table from the previous slide is known as a **confusion matrix**
AKA **contingency table**:

		Predicted condition	
		Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

Note that the *positive class* is determined by the data analyst

Receiver Operating Characteristics (ROC) Analysis

Receiver Operating Characteristics (ROC) reflect the *row proportions* of the confusion matrix:

- ▶ **True positive rate** (TPR), or $\frac{\text{True Positives}}{\text{Total Positives}}$, also known as *sensitivity*, *hit rate*, and *recall*
- ▶ **False positive rate** (FPR), or $\frac{\text{False Positives}}{\text{Total Negatives}}$, also known as $1 - \text{specificity}$

A perfect classifier has a TPR of 1 and a FPR of 0, but there's an inherent trade off between the two quantities.

Receiver Operating Characteristics (ROC) Analysis

- ▶ Classification algorithms return a *score*, or a predicted probability $\hat{p}$ of the positive class
 - ▶ These scores must be mapped to a class label $\hat{y} = 1$, with the typical threshold p^* for binary classification being 0.5
i.e. predict $\hat{y} = 1$ when $\hat{p} \geq p^*$
 - ▶ This threshold p^* can be changed to manipulate the trade off between TPR and FPR

Receiver Operating Characteristics (ROC) Analysis

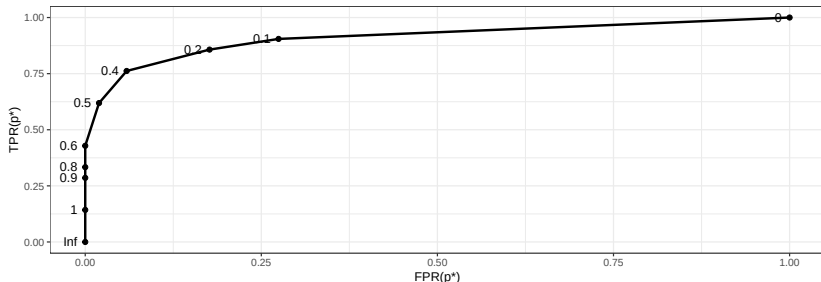
- ▶ Classification algorithms return a score, or a predicted probability $\hat{p}$ of the positive class
 - ▶ These scores must be mapped to a class label $\hat{y} = 1$, with the typical threshold p^* for binary classification being 0.5 i.e. predict $\hat{y} = 1$ when $\hat{p} \geq p^*$
 - ▶ This threshold p^* can be changed to manipulate the trade off between TPR and FPR
- ▶ For example, a “positive” classification for any observation with at least 3 of 8 neighbors having cancer leads to the following confusion matrix:

	Predicted Cancer	Predicted Healthy
Has Cervical Cancer	16	5
Doesn't Have Cancer	3	48

- ▶ The TPR is now 76.2% (compared to 61.9%), but the FPR is 5.9% (compared to 2.0%)

ROC Curves

The trade-off between $\text{TPR}(p^*)$ and $\text{FPR}(p^*)$ of a classifier can be summarized by an ROC curve where the dots 0, 0.1, 0.2, $\dots$, 1 represent all possible values of p^* :



The *area under the ROC curve* (AUC) provides a single-number summary of how well the classifier performs. An AUC value of 0.5 reflects no predictive value, while 1.0 indicates perfect classification.

Other Performance Metrics Based on Confusion Matrices

There are many other metrics that can be calculated from a confusion matrix ([Wikipedia Link](#))

SKIP: Class Imbalances

- ▶ A strength of AUC is that it is not influenced by imbalances in the class distribution
 - ▶ In our example, 70.8% of the sample did not have cancer, so we could achieve 70.8% classification accuracy *without the model adding any predictive value*

SKIP: Class Imbalances

- ▶ A strength of AUC is that it is not influenced by imbalances in the class distribution
 - ▶ In our example, 70.8% of the sample did not have cancer, so we could achieve 70.8% classification accuracy *without the model adding any predictive value*
- ▶ As a more extreme example, research by the St Louis federal reserve found the delinquency rate across all US loans to be 1.2%
 - ▶ In this application, would you be impressed by an algorithm that correctly classifies loan delinquency with 99% accuracy?

SKIP: Balanced Accuracy

- ▶ **Balanced accuracy** addresses class imbalance by averaging the accuracy within each class (ie: averaging the accuracy achieved in each row of the confusion matrix).
- ▶ For binary classification, this amounts to:

$$\text{Balanced Accuracy} = \frac{\text{TPR} + (1 - \text{FPR})}{2}$$

- ▶ Note that this is the average of the TPR and the *true negative rate*.

SKIP: Precision and Recall

- ▶ TPR and FPR both address the question “how many of each observed class are classified as positive?”
 - ▶ **Precision** answers the question “how many of the samples that were classified as positive are actually positive?”
 - ▶ **Recall** is often analyzed in conjunction with precision, but it's just another term for TPR

$$\text{Precision} = \frac{\text{True Positives}}{\text{Total Predicted Positives}}$$

$$\text{Recall} = \text{TPR} = \frac{\text{True Positives}}{\text{Total Positives}}$$

SKIP: F1 Score

The **F1 Score** is a popular metric that combines a classifier's precision and recall by taking their **harmonic mean**:

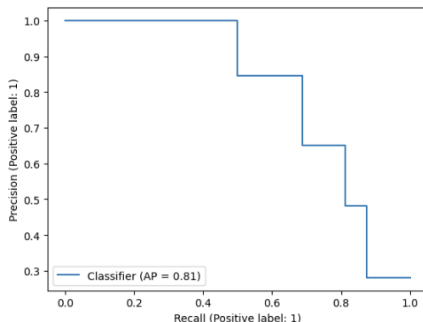
$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Since F1 focuses on the prediction of the positive class, it tends to be popular in applications that seek to identify relatively uncommon events.

- ▶ Examples include: predicting loan default, classifying spam emails, etc.

SKIP: PR-AUC

Precision and recall can also be explored at a variety of classification thresholds in a PR curve:



The area under this curve, known by *PR-AUC* or *AP*, is an alternative single number summary of a classifier's performance.

SKIP: Multiple Classes

Extensions to multi-label classification requires adopting *one-vs-rest scheme* (combining many classes to form the “negative” class) or a *one-vs-one scheme* (pairwise comparisons)

	Pr Setosa	Pr Versicolor	Pr Virginica
Setosa	50	0	0
Versicolor	0	48	4
Virginica	0	2	46

- ▶ Under a one-vs-many scheme:
 - ▶ For Versicolor flowers, the TPR is $48/52$ and the FPR is $2/98$
- ▶ Under a one-vs-one scheme:
 - ▶ For Versicolor flowers compared to Virginica flowers, the TPR is $48/52$ but the FPR is $2/48$

SKIP: Micro vs. Macro Averaging with Multiple Classes

For multi-label applications there are two popular for calculating single number metrics like AUC or the F1-score:

1. *Micro-averaging* - aggregate the contributions from each class when calculating the metric (only applicable in the one-vs-many scheme)
2. *Macro-averaging* - calculate the metric independently for each class, then take the average (applicable for both one-vs-many and one-vs-one schemes)

For the Iris example:

1. The *micro-averaged TPR* is $\frac{\sum_{j=1}^k \text{True Pos.}}{\sum_{j=1}^k \text{Total Pos.}} = \frac{50+48+46}{50+52+48} = 0.9600$
2. The *macro-averaged TPR* is $\frac{1}{k} \sum_{j=1}^k \text{TPR}_j = \frac{(50/50)+(48/52)+(46/48)}{3} = 0.9605$

Which Metric(s)?

Consider the following confusion matrix:

	Pos	Neg
Pos	10	10
Neg	5	995

- ▶ Classification accuracy is 98.5%
- ▶ Balanced accuracy is 74.75%
- ▶ The F1-score is 0.571

Which metric provides the most useful assessment of this classifier?

Recommendations

- ▶ Classification accuracy is acceptable when classes are roughly balanced, and false positives/negatives are equally problematic
 - ▶ It is also the most easily interpreted metric, so non-technical clients may prefer it
- ▶ ROC analysis and AUC are useful when classes are imbalanced, and false positives/negatives have differential impact
- ▶ SKIP: Balanced accuracy is useful when classes are imbalanced, and false positives/negatives are equally problematic
- ▶ SKIP: The F1-score and PR analysis are useful when classes are imbalanced, and you mostly care about predicting the positive class